

JINJIE WANG

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HIGHLIGHTS

- Completed multiple advanced AI elective courses and gained a deep understanding of the operating principles of various AI technologies, such as decision trees, deep learning, clustering and heuristic search
- Developed expertise in Robot Operating System (ROS) and Arduino, focusing on control and vision applications through hands-on project experience
- Participated in Nanyang Technological University's GEM Trailblazer Summer 2023 programme, achieving an A+ in the module, *Artificial Intelligence and Data Mining*
- Gained professional experience as a Software Engineer and Database Development Intern, developing AI-based systems and WeChat mini-programs using microservice architectures, FastAPI, Next.js, PostgreSQL, as well as conducting system testing and user needs analysis

EDUCATION

City University of Hong Kong (CityU)

Hong Kong, China | 9/2022 - 6/2026

Bachelor of Science in Computer Science

Relevant courses: *Artificial Intelligence and Data Mining (A+)*, *AI Game Programming (B+)*, *Computer Vision and Image Processing (B)*, *Fundamentals of Data Science (B+)*, *Software Engineering Practice (A)*, *Computer Programming (A+)*, *Design and Analysis of Algorithms (B+)*

GPA: 3.43/4.30

ACADEMIC PROJECTS

International Collegiate Programming Contest (ICPC)

Shenyang and Nanjing, China | 11/2023 - 11/2023

Core Team Member

- Specialised in graph theory and optimisation, mastering fundamental algorithms, such as simulation, recursion, divide-and-conquer, dynamic programming, binary search, greedy and data structures
- Implemented advanced graph algorithms, including heavy-light decomposition, union-find, minimum spanning tree, shortest path algorithms (e.g., Dijkstra, SPFA), tree diameter and differential constraints to address dynamic pathfinding and network optimisation problems
- Achieved efficient code optimisation and collaborative problem-solving strategies

JUNEER Fall Training of CityU Underwater Robotics

Hong Kong, China | 9/2023 - 12/2023

Core Team Member

- Directed hardware-software integration for submersible robots, including thruster control systems and sensor calibration
- Programmed real-time embedded control logic in C++ on the Arduino platform and configured power distribution circuits for depth stability
- Coordinated tank testing sessions to validate hydrodynamic performance and sensor feedback reliability

Artificial Intelligence and Data Mining Coursework

Jurong, Singapore | 7/2023 - 8/2023

Independent Contributor

- Developed a neural network-based housing price prediction model using Python with TensorFlow and Scikit-learn, incorporating geographic and economic feature engineering
- Implemented data preprocessing pipelines to handle missing values and normalise heterogeneous real estate datasets

Enhancing Your Service Leadership for the 21st Century Group task

Hong Kong, China | 1/2024 - 5/2024

Core Team Member

- Designed a wheelchair accessibility detection system using YOLOv5 trained on custom-labelled urban scene datasets
- Proposed AI-assisted infrastructure audit framework, emphasising inclusive public space design

AI Game Programming Coursework

Hong Kong, China | 1/2025 - 5/2025

Core Team Member

- Engineered adversarial game AIs for Gomoku and Tessellation using Monte Carlo Tree Search (MCTS) with heuristic evaluation functions
- Optimised computational efficiency via alpha-beta pruning and parallel simulation for complex board states

Grand Prix Metaverse Autonomous Driving Challenge 2023

Hong Kong, China | 10/2023 - 12/2023

Core Team Member

- Optimised perception-control pipeline parameters for championship-winning autonomous racing models, using PID for longitudinal control and MPC for lateral control
- Implemented Bayesian optimisation and Machine Learning to refine trajectory planning in simulated high-speed cornering scenarios
- Self-learned and applied ROS

PROFESSIONAL EXPERIENCES

Nanjing Liandi Information Systems Co., Ltd. Shanghai Branch

Shanghai, China | 7/2025 - 8/2025

Software Development Engineer Intern

- Developed an AI-based meeting record processing system for audio transcription, summary generation and knowledge base management, utilising a microservice architecture with the Dify workflow engine, FastAPI, SQLAlchemy for the backend, Next.js for the frontend and PostgreSQL as the primary database
- Implemented and integrated APIs for locally deployed Whisper and PyAnnote models
- Created test cases applying condition coverage or path coverage, depending on code complexity
- Gained hands-on experience with the structure, version control and development models of enterprise-level software projects
- Performed development on Ubuntu 22.04 via SSH, mastering essential operations and configuring development environments and dependencies

Spark Technology Innovation Limited

Hong Kong, China | 8/2024 - 5/2025

Software Development Engineer Intern

- Explored the company's business processes and conducted on-site investigations into user needs
- Cooperated with a colleague to develop a WeChat mini-program for warehouse management, featuring warehousing/ex-warehousing, historical records, permission management, inventory checks and data analysis; took primary responsibility for backend systems and database architecture
- Conducted system testing, wrote user manuals and maintained post-launch updates and bug fixes

AWARDS

- Dean's list at CityU in 2023
- Champion at the Grand Prix Metaverse Autonomous Driving Challenge in 2023

ADDITIONAL SKILLS

IT Skills: C++, Python, HTML, CSS, JavaScript, Java, Ubuntu, Microsoft Office

Languages

Mandarin Chinese: Native; English: Fluent (IELTS: 7.5)